CNVS Formal Mathematical Synthesis of the Extended Theory

Versione 5.5

This document constitutes the compact mathematical formalization of the complete theory presented in The Theory of Closed Native Verification Systems (CNVS) v5.5, preserving its core axioms, definitions, and security theorems in a concise and self-contained form.

General Index

Abstract

This paper introduces closed native verification systems (CNVS), a formal class of verification systems in which state membership is verification-dependent rather than primitive. The theory combines a typed structural universe, a type-preserving recursive decomposition, random assignment of non-uniform fragments, limited verifier knowledge, and global invariant checking, first enforcing data convergence in a local verification environment.

External verifiers are not provided with the internal declared payload D_i; They receive only a task specification and independently measure the assigned attribute, the result of which is internally compared to the value maintained by the system.

CNVS therefore does not perform syntactic slicing of the payload. The decomposition generates atomic evaluation primitives (which have the local semantics to be computable via a Local Verification function) and injects them into a randomized routing topology.

The fragment is not the payload, but a measurement grid that is randomly distributed along the graph and that requires a convergent observation with respect to the unknown payload.

Lemma 10 ensures that the shape of this routing network does not mimic the original information tree, preventing metadata leakage.

A conditional probabilistic theorem limits unauthorized reconstruction through weighted adversarial information and exponential concentration inequalities. An asymptotic scaling theorem shows that security can increase as the system expands in the presence of explicit assumptions.

The technical difficulty of reconstruction outside the system arises from the strictly local knowledge available to each verifier.

1. Primitive Domains and Typed Structural Universe

Let $B = \{0, 1\}$ and let B^* be the set of all finite binary strings.
Let $\mathcal{T}$ be a strictly finite set of structural types.

To guarantee semantic consistency and type preservation, we define the primitive domains not as flat sets, but as families of sets indexed by their structural type $\sigma \in \mathcal{T}$

- $Id_\sigma \subseteq B^*$: the space of identifiers for type σ .
- $D_\sigma \subseteq B^*$: the space of declared data payloads structured according to type σ .
- $At_\sigma \subseteq B^*$: the space of attribute classes for type σ .

- $Obs_\sigma \subseteq B^*$: the space of externally reported values for type σ .

Let P be a finite set of logical properties, and let $\mathcal{P}(P)$ denote the set of all its strictly finite subsets.

To model the absence of data without introducing domain collisions or ambiguity with the empty string $\varepsilon \in B^*$, we introduce a distinguished element $\perp \notin B^*$. We define the observational domain as a disjoint union:

$Obs^\perp_\sigma \triangleq Obs_\sigma \uplus \{\perp\}$

where the $\perp$ element rigorously denotes the absence of an external observation at non-terminal structural levels.

We define the Structural Universe $\boldsymbol{\mathcal{S}}$ as a dependent sum (Sigma-type) over the set of structural types $\boldsymbol{\mathcal{T}}$:

(1)

$\boldsymbol{\mathcal{S}} \triangleq \Sigma_{\sigma \in \mathcal{T}} (Id_\sigma \times D_\sigma \times At_\sigma \times Obs^\perp_\sigma \times \mathcal{P}(P))$

Consequently, each structural state $s \in \boldsymbol{\mathcal{S}}$ assumes the dependent form:

$s = \langle \sigma. (id, d, a, obs, p) \rangle$

This construction guarantees ontologically that the identifier, the payload, and the attributes are intrinsically bound to their specific structural type t at instantiation, preventing type-mismatch vulnerabilities prior to any verification phase.

2. State Space and Verification Signatures

A system state at time t is formally defined as a tuple:

(2)

$S(t) = \langle E(t), R(t), C(t) \rangle$

where:

- $E(t) \in \mathcal{P}_{fin}(\boldsymbol{\mathcal{S}})$ is a finite set of structural elements active in the system at time t.
- $R(t) \subseteq E(t) \times E(t)$ is the relational graph defining structural dependencies between elements.
- $C = \{c_1, ..., c_m\}$ is a finite set of global system invariants (consistency constraints). Formally, let $\mathcal{C}$ be the universe of all definable structural predicates. Each invariant

$c_i \in \mathcal{C}$ is a boolean function $c_i : \mathcal{P}_{fin}(\mathcal{S}) \times \mathcal{P}_{fin}(\mathcal{S} \times \mathcal{S}) \to \{0, 1\}$ that evaluates a specific topological or logical consistency rule over the elements and their relations.

The State Space Σ is therefore the strictly bounded set of all such well-formed tuples:

$\Sigma \triangleq \{\langle E(t), R(t), C(t)\rangle \mid E \in \mathcal{P}_{fin}(\mathcal{S}), R \subseteq E \times E, C \subseteq \mathcal{C}\}$

Terminal Elements (Ter)

Let $D: S \to \mathcal{P}_{fin}(S)$ be the recursive decomposition operator. We formally define the subset of terminal fragments (leaf nodes) at time t as the elements that cannot be further decomposed:

$Ter(t) \triangleq \{s \in E(t) \mid D(s) = \emptyset\}$

Hierarchical Depth (n) and Partial States

Because the decomposition operator D is recursively applied, we define the hierarchical depth of the system as a natural number $n \in \mathbb{N}_{>0}$, representing the maximum path length from any root element to its terminal leaves. Consequently, the state S(t) can be stratified into n discrete topological levels. We denote $S^{(k)}(t)$ as the partial state reconstruction at level k, where $k \in \{0, ..., n-1\}$. By definition, the base level $S^{(0)}(t)$ contains the terminal fragments Ter(t), while the maximal level $S^{(n-1)}(t)$ corresponds to the fully reconstructed roots.

Definition of Relational Consistency (Cons_R)

To ensure that the state topology strictly adheres to the decomposition rules, the relational consistency predicate evaluates to 1 if and only if the edge set R(t) perfectly maps to the recursive decomposition operator D:

$Cons_R(R(t)) = 1 \Leftrightarrow (\forall(s, s') \in E(t) \times E(t), [(s, s') \in R(t) \Leftrightarrow s' \in D(s)]$

Verification Signatures

The Local Verification Function V_L evaluates the formal admissibility and attribute convergence of a single element. It maps an element to a tuple containing a boolean validity flag and a finite set of verified logical properties P:

$V_L : \mathcal{S} \to \{0, 1\} \times \mathcal{P}(P).$

Let $\pi_1 : \{0, 1\} \times \mathcal{P}(P),$
be the standard projection operator extracting the boolean verification result.

The Global Verification Function V_G acts as the absolute structural veto for the framework. It evaluates the entire system state $S(t) = \langle E(t), R(t), C(t)\rangle$, mapping it to a boolean validity flag:

$V_G : \Sigma \to \{0, 1\}$

Using the predefined hierarchical depth n, the preliminary global validity condition evaluates the system recursively from the terminal leaves up to the root:

$V_G(S(t)) = 1 \Leftrightarrow (\forall s \in Ter(t), \pi_1(V_L(s)) = 1) \wedge Cons_R(R(t)) \wedge (\forall c_i \in C, c_i\,(E(t), R(t)) = 1)$.

This preliminary recursive schema is formalized and strictly bounded in Axiom III, where global verification is definitively decomposed into local admissibility, relational consistency, and invariant satisfaction.

3. Local Verification and Admissibility Signatures

To formalize the local evaluation of a structural element, we define the verification primitives directly over the type-indexed components of the Structural Universe $\boldsymbol{\mathcal{S}}$.

Let the Local Verification Function V_L be formally declared with the following signature:

$V_L : \boldsymbol{\mathcal{S}} \rightarrow \{0, 1\} \times \boldsymbol{\mathcal{P}}(P)$,

for each structural type $t \in \boldsymbol{\mathcal{T}}$, we define two distinct boolean verification predicates operating strictly on their corresponding index-bound domains:

1. Formal Admissibility ($Form_\sigma$): Evaluates the structural integrity, formatting, and cryptographic bounds of the element's components.

   $Form_\sigma : Id_\sigma \times D_\sigma , \times At_\sigma \times Obs^{\perp}{}_\sigma \rightarrow \{0, 1\}$

2. Attribute Convergence ($Conv_\sigma$): Evaluates the semantic coherence between the declared payload d, the governing attributes a, and the externally reported values obs.

   $Conv_\sigma : At_\sigma \times D_\sigma \times Obs^{\perp}{}_\sigma \rightarrow \{0, 1\}$.

To guarantee decidability and consistency for internal structural elements lacking physical external observations, we formally impose the following axiomatic condition for the vacuous truth of convergence:

$\forall a \in At_\sigma , \forall d \in D_\sigma , \quad Conv_\sigma(a, d, \perp) = 1$

Given an element $s = \langle \sigma, (id, d, a, obs, p) \rangle \in \boldsymbol{\mathcal{S}}$, the operational definition of V_L is therefore constructed as:

(4)

$V_L(s) \triangleq \langle Form_\sigma(id, d, a, obs) \wedge Conv_\sigma(a, d, obs), p \rangle$

where $\wedge$ denotes the strict logical AND operator.

Let $\pi_1 : \{0, 1\} \times \mathcal{P}(P) \to \{0, 1\}$ be the standard projection to the boolean domain.
We define the local admissibility condition Adm_L for an element s as:

$Adm_L(s) = 1 \iff \pi_1(V_L(s)) = 1$

This architecture ensures that verification is inherently type-preserving and free of domain collisions, satisfying the prerequisites for the global state evaluation.

3a. Epistemic Restriction in Local Verification

In CNVS, the External Verification Agent does not receive the internally declared datum d associated with the Fragment. Instead, the Agent independently observes or measures a local property of the Object under restricted knowledge conditions.
In the evaluation of $V_L(s)$ for any terminal element $s = \langle \sigma, (id, d, a, obs, p) \rangle \in \mathcal{S}$, the internal payload $d \in D_\sigma$ is strictly restricted to the CNVS system.
The external verification agent assigned to element s receives only the task specification derived from $a \in At_\sigma$ and independently produces the observed value $obs \in Obs^{\perp}_{\sigma}$. The system natively evaluates $Conv_\sigma(a, d, obs)$ without external payload disclosure.

Remark 3a. (Polymorphism of Convergence and Tolerance Bounds)

The Local Verification Function abstracts the physical and semantic complexity of real-world observations by delegating the tolerance logic to the attribute domain At_σ. The convergence predicate $Conv_\sigma$ acts as a polymorphic interface that accommodates both continuous and discrete state evaluations, ensuring that minor environmental noise or subjective categorizations do not break the systemic decidability.

Depending on the structural type $\sigma \in \mathcal{T}$, the evaluation falls into two macro-categories, governed entirely by the parameter $a \in At_\sigma$:

1. Quantitative Convergence (Metric Spaces):

   If σ dictates a measurable physical scalar or vector space, the attribute a securely encodes a tolerance threshold ε_a. The convergence evaluates to 1 if and only if the distance function $\boldsymbol{\delta}_\sigma$ between the payload d and the observation obs satisfies the bound:

   $Conv_\sigma(a, d, obs) = 1 \Leftrightarrow \boldsymbol{\delta}_\sigma(d, obs) \leq \varepsilon_a$

2. Qualitative Convergence (Categorical and Semantic Spaces):

If σ represents a qualitative, logical, or symbolic state (e.g., colors, classification labels, bit-strings), the tolerance is modeled as a semantic neighborhood or an equivalence class mapping. The attribute a defines an acceptable boundary function $\Psi_a : D_\sigma \rightarrow \mathcal{P}(Obs_\sigma)$, which maps the expected datum to a finite set of admissible observations (e.g., mapping "red" to a cluster of acceptable hex-color subsets or semantic neighbors). Convergence evaluates to 1 if and only if the observation falls within this valid neighborhood:

$Conv_\sigma(a, d, obs) = 1 \Leftrightarrow obs \in \Psi_a(d)$

This architecture securely delegates the specific digitalization encodings (e.g., continuous-to-discrete quantization, threshold algorithms, clustering) to the system implementation layer, while maintaining a mathematically strictly bounded global verification.

4. Recursive Decomposition and Reconstruction

Let type : $\mathcal{S} \rightarrow \mathcal{T}$ be the fundamental projection function that assigns to each structural element its specific structural type.

We define the primary decomposition operator D as a function mapping a well-formed element to a strictly finite set of its constituent sub-elements:

$D : \mathcal{S} \rightarrow \mathcal{P}_{fin}(\mathcal{S})$.

To ensure logical consistency and non-trivial state distribution, we impose a strict disjunctive cardinality constraint on the output of D(s) for any element $s \in \mathcal{S}$:

$|D(s)| = 0 \lor |D(s)| \geq 2$

if $|D(s)| = 0$(i.e., $D(s) = \emptyset$), the element cannot be further decomposed and is formally classified as a terminal node ($s \in Ter$).
If $|D(s)| \geq 2$, the element is decomposed into a finite set of child fragments $\{f_1, f_2, \ldots., f_k\}$.

To guarantee structural consistency across hierarchical levels, the decomposition operator must be strictly type-preserving. For any element and its resulting fragments, the structural type remains invariant:

(6)

$\forall s \in \mathcal{S}, \forall f_i \in D(s), \quad type(f_i) = type(s)$.

Conversely, we define the formal reconstruction operator Rec. To prevent illicit totality (i.e., mapping arbitrary or malicious subsets to valid parent elements), Rec is explicitly defined as a partial function operating exclusively over the exact image of the decomposition operator:

$\mathrm{Rec} : \mathrm{Image}(D) \setminus \{\emptyset\} \rightharpoonup \mathcal{S}$

The framework mandates perfect structural left-inversion, ensuring that for any valid decomposition: (7)

$\forall s \in \mathcal{S}$, such that $|D(s)| \geq 2$, $\mathrm{Rec}(D(s)) = s$

under deterministic canonical serialization.

5. Non-Uniform Fragmentation

Let $I_{total} : \mathcal{S} \to \mathbb{N}$ denote the total informational complexity (measured in bits) of any structural element $\forall s \in \mathcal{S}$. We formally decompose this measure into two strictly positive functions:

- $I_{payload} : \mathcal{S} \to \mathbb{N}$ defining the semantic informational content inherited from the original root element.
- $I_{metadata} : \mathcal{S} \to \mathbb{N}$ defining the systemic structural information required for fragment identification, assignment consistency, relational reconstruction, and verification coordination.

For every structured element $s \in \mathcal{S}$, the total encoded information is defined as the linear sum:

(8)

$I_{total}(s) = I_{payload}(s) + I_{metadata}(s)$.

Remark 5.

The CNVS framework strictly distinguishes between semantic payload ($I_{payload}$) and systemic coordination data ($I_{metadata}$). Unauthorized reconstruction is therefore bounded not solely by the entropic threshold of the semantic fragments, but by the computational hardness of inferring the structural graph topology, relational dependencies, and admissible invariants encoded within $I_{metadata}$.

Non-Uniformity Axiom.

To mathematically enforce this informational asymmetry, the recursive decomposition D is intentionally non-uniform. For any non-terminal element $\forall s \in \boldsymbol{\mathcal{S}}$ such that $D(s) \neq \emptyset$, there exist at least two child fragments within its decomposition that carry strictly unequal semantic payload measures:

1. $\forall s \in \boldsymbol{\mathcal{S}}$ such that $D(s) \neq \emptyset$, $I_{payload}(s) = \Sigma_{fi \in D(s)}\ I_{payload}(f_i)$,

(9)

2. $\forall s \in \boldsymbol{\mathcal{S}}$ such that $D(s) \neq \emptyset$, $\exists\ f_a, f_b \in D(s)$ such that $I_{payload}(f_a) \neq I_{payload}(f_b)$

Practical Decomposition Limit.

Recursive decomposition cannot grow indefinitely without incurring a monotonic increase in structural cost. Excessive fragmentation linearly increases the metadata overhead ($I_{metadata}$) while reducing the operational significance of the payload ($I_{payload}$) carried by terminal fragments. Consequently, the CNVS framework admits a strictly finite decomposition limit governed by this informational balance, mathematically averting asymptotic non-termination anomalies.

6. Knowledge Restriction and Asymptotic Negligibility

To formalize the Zero-Knowledge and Multi-Party Computation properties of the framework, we must strictly bound the informational exposure to external actors.

Let $\mathcal{V}$ be a finite set of external verifiers. We introduce the knowledge measure function $\mathcal{K} : \mathcal{V} \times \boldsymbol{\mathcal{S}} \rightarrow \mathbb{N}$, which quantifies the exact structural and semantic information (in bits) exposed to a specific verifier $v \in \mathcal{V}$ concerning an element $s \in \boldsymbol{\mathcal{S}}$ during the verification process.

For a verifier $v \in \mathcal{V}$ assigned to validate a specific attribute $a \in At_\sigma$ of a terminal fragment $f \in Ter(t)$, let $I_{min}(a) \in \mathbb{N}_{>0}$ denote the absolute minimum informational complexity required to perform the assigned task without semantic loss. Knowledge cannot converge to zero absolutely ($\mathcal{K}(v, f) \geq 1$), as a complete loss of information would render the verification undecidable.

We formally define the structural domains of total information iteratively:

- $I_{total}(f)$: The total information encoded in the terminal fragment f.
- $I_{total}(s)$: The total information encoded in the reconstructed parent node $s = Rec(D(s))$.

- $I_{global}(S(t))$: The absolute total information of the entire system state at time t.

The framework imposes the following Strict Informational Subordination Chain:

(10)

$$I_{min}(a) \le \mathcal{K}(v,f) < I_{total}(f) \ll I_{total}(s) \ll I_{global}(S(t))$$

Remark 6.

In general, the verifier's accessible knowledge $\mathcal{K}(v,f)$ does not coincide with the fragmental information $I_{total}(f)$ itself. The internal declared payload $d \in D_\sigma$ remains strictly confined within the system boundary. The verifier receives only a minimal semantic projection required to independently report an external observation $obs \in Obs^{\perp}{}_{\sigma}$.

The strict inequality $\mathcal{K}(v,f) < I_{total}(f)$ guarantees that no single verifier holds the complete semantic state of a fragment.

Furthermore, the operator $\ll$ denotes asymptotic informational negligibility.
To formalize this without violating the finite cardinality bounds of $\boldsymbol{\mathcal{P}}_{fin}(\boldsymbol{\mathcal{S}})$, we define a sequence of system configurations parameterized by a global structural fragmentation scale $m \in \mathbb{N}$, defined as the lower bound of system-wide decomposition: $m = \min_{s \notin Ter(t)} |D(s)|$.

By the principle of informational conservation, the parent node information scales with its partition complexity: $I_{total}(s) \ge \Sigma_{f\_i \in D(s)} I_{total}(f_i)$.
Consequently, for any strictly bounded local verifier knowledge $\mathcal{K}(v,f)$, the informational ratio converges to zero as the macroscopic system scale expands:

$$\lim_{m \to \infty} \sup_{v \in \mathcal{V}, s \notin Ter(t), f \in D(s)} (\mathcal{K}(v,f) / I_{total}(s)) = 0$$

This guarantees that local knowledge restriction scales asymptotically into global cryptographic security.

Remark 6. Definition: Structural Nature of Terminal Fragments

To preclude fundamental misinterpretations of the Knowledge Restriction limits, it is strictly necessary to define the structural nature of a fragment within the CNVS framework.

A terminal fragment $f \in Ter(t)$ is not a syntactic fraction (e.g., a data "shard" or a cryptographic sub-string) of the global payload.

Instead, f is formally defined as an Autonomous Evaluative Primitive. The recursive decomposition function D_t does not slice the data geometrically; it extracts localized logical conditions from the global state and encapsulates them into discrete, executable measurement tasks.

Consequently, the terminal fragment possesses perfect local semantic coherence — satisfying the semantic lower bound $I_{payload}(f) \geq I_{min}$ — which enables the external verifier to execute the task and yield a deterministic Boolean output. However, because this localized task is structurally decoupled from the global relational graph, the verifier extracts no semantic knowledge regarding the overarching systemic payload.

In conclusion, in CNVS, the system does not destroy the semantic content, but extracts the logical predicates (the truth conditions) and transforms them into Local Tasks that are non-uniform in content and volume (Terminal Fragments) which are then randomly distributed along a hidden graph to disperse the connections.

7. Protocol Parameters

To govern the informational density, fragmentation topology, and semantic granularity of the system, the CNVS framework establishes a set of strictly typed control parameters:

- $\epsilon_G \in (0,1]$: Global Information Density Threshold.
- $\epsilon_N \in (0,1]$: Nodal Information Density Threshold.
- $\epsilon_C \in (0,1]$: Composite Information Density Threshold.
- $\lambda \in \mathbb{R}_{>0}$: Fragmentation-control parameter regulating decomposition depth, redundancy, and structural distribution.
- $I_0 \in \mathbb{N}_{>0}$: Minimum semantic informational quantum admissible within the System.

8. Information Density

We quantify the relative informational weight of any terminal fragment $f \in Ter(t)$ with respect to its parent node $s \in \boldsymbol{S}$ (where $f \in D(s)$) and the total system state $S(t)$.

Using the established informational measures I, we define the dimensionless density ratios $\rho \in [0, 1] \subset \mathbb{R}$.

The Nodal Information Density evaluates the fragment's weight relative to its immediate structural root:

(11)

$$\rho_N(f,s) = I_{total}(f) / I_{total}(s)$$

(12)

$\rho_G(f,S(t)) = I_{total}(f) / I_{global}(S(t))$

(13)

$\rho_C(f,s,S(t)) = \rho_N(f,s)^{\alpha} \quad \rho_G(f,S(t))^{\beta}$

where α , $\beta \in [0,1]$ and $\alpha + \beta = 1$

The scalar quantity ρ_C dictates the composite topological relevance of the fragment. During the admissibility check, these deterministic evaluations are matched against the axiomatic thresholds (ϵ_G, ϵ_N, ϵ_C). Any partial state mapping to a density tensor that violates these constraints is inherently rejected by the Global Veto, preventing structural starvation or monopolization.

9. Random Assignment

To prevent deterministic collusion and isolate adversarial clusters, terminal fragments are dynamically assigned to external verifiers via a randomized topological mapping.

Let $\mathcal{V}$ be the finite set of external verifiers, and let Ξ denote a uniform cryptographic probability space (source of secure randomness). The assignment mapping A_t at time t is formalized as a probabilistic function:

(14)

$A_t : Ter(t) \times \Xi \rightarrow \mathcal{V}$

To guarantee the applicability of concentration inequalities, the assignment of any terminal fragment f_i is strictly statistically independent of the assignment of any other fragment f_j. Thus, the random variables derived from A_t are Independent and Identically Distributed (I.I.D.) over $\mathcal{V}$.

10. Axiom I: Verification-Independent Existence

The existence of an element s within the system state S(t) is independent of its verification status.

The verification process is a mapping $V_G : \Sigma \to \{0, 1\}$ **(15)** that partitions Σ into valid and invalid states, without altering the membership of elements within E(t).
Thus, $V_G(S(t)) = 1$ is a condition for admissibility, not existence.

11. Axiom II: Randomized Non-Uniform Fragmentation

Every element $s \in \boldsymbol{S}$ such that $D(s) \neq \emptyset$ is recursively decomposed into a non-uniform set of fragments. The assignment operator A_t defines a probabilistic mapping to the verifier set $\mathcal{V}$:

- $|D(s)| \geq 2$
- $\exists\, f_a, f_b \in D(s) : I_{payload}(f_a) \neq I_{payload}(f_b)$
- $A_t : Ter(t) \times \Xi \to \mathcal{V}$
- $Rec(D(s)) = s$

12. Axiom III: Non-Reducibility of Global Validity

Global validity is not an emergent property of terminal elements, but a systemic invariant. Let $Cons_R(R(t), E(t))$ be the relational consistency and $Inv_C(S(t))$ be the satisfaction of the global invariants C. The Global Verification Function is defined as the strict conjunction: (16)

$V_G(S(t)) = 1 \Leftrightarrow (\forall s \in Ter(t), \pi_1(V_L(s)) = 1) \wedge Cons_R(R(t), E(t)) \wedge Inv_C(S(t))$ **(16)**

Non-Reducibility Principle:

$\exists S(t) \in \Sigma$ such that $(\forall s \in Ter(t), \pi_1(V_L(s)) = 1) \wedge (V_G(S(t)) = 0)$ **(16a)**

This asserts the existence of "locally valid but globally inconsistent" states. This proves that the global system state cannot be reduced to the mere verification of its terminal fragments, thereby mandating the Global Veto.

12a. State Transition Law

The evolution of the system state to a candidate successor S' is governed by the absolute admissibility condition:

(16c)

$$S(t+1)=\begin{cases} S' & \text{if} \quad V_G(S')=1 \\ S(t) & \text{if} \quad V_G(S')=0 \end{cases}$$

Any candidate state S' such that $V_G(S') = 0$ is formally rejected.
This establishes that state evolution is strictly verification-dependent and constrained by the totality of structural invariants C and the relational consistency $Cons_R$. The system admits no "invalid" transitions; the state space is self-correcting via the Global Veto.

13. Fundamental Lemmas

The robustness of the CNVS framework is supported by the following lemmas, derived from the axioms of structural decomposition and local verification:

- **Lemma 1 (Terminal Membership):** $\forall s \in Ter(t), s \in E(t) \subseteq \boldsymbol{\mathcal{S}}$ (Existence is primitive).
- **Lemma 2 (Global Necessity):** $\exists S(t) \in \Sigma : (\forall s \in Ter(t), Adm_L(s)=1) \wedge V_G(S(t))=0$ (Proves that local validity does not imply global consistency).
- **Lemma 3 (Type Invariance):** $\forall \sigma \in \mathcal{T}, \forall s \in D_\sigma \Longrightarrow (\forall f \in D(s), f \in D_\sigma)$
- **Lemma 4 (Payload Conservation):** $\forall s \in \boldsymbol{\mathcal{S}}, I_{payload}(s) = \Sigma_{f\_i \in D(s)} I_{payload}(f_i)$ (Semantic content is conserved across hierarchies).
- **Lemma 5 (Metadata Overhead):** $\forall s \in \boldsymbol{\mathcal{S}}, I_{metadata}(Rec(D(s))) < \Sigma_{f\_i \in D(s)} I_{metadata}(f_i)$. (Fragmentation increases systemic metadata costs).
- **Lemma 6 (Asymptotic Density Decay):** $\lim_{m \to \infty} \rho_N(f, s) = 0$ (Under arbitrary fragmentation, the relative density of a terminal fragment vanishes).

- **Lemma 7 (Structural Inference Bound):** see chapter 18.a.
- **Lemma 8 (Topological Underdetermination Bound):**

  To preclude the vulnerability of deterministic algebraic reconstruction via global invariant equations, the structural decomposition of the CNVS mathematically guarantees strict algebraic underdetermination.

  Let C_{eq} denote the finite set of algebraic or logical equations imposed by the Global Invariants (C) on a state S.
  Let $\mathbb{F}_{miss} \subset \mathrm{Ter}(t)$ be the set of uncompromised critical fragments required for unauthorized reconstruction.

  For any adversarial coalition attempting to infer $\mathbb{F}_{miss}$ utilizing the known compromised subset and the global invariant rules C_{eq}, the total number of informational unknowns N_u is defined not merely by the missing semantic payload, but inherently by the missing relational topology encoded within the cryptographic space Ξ.

  Formally, the dimensionality of the unknowns is:

  $N_u = \dim(I_{payload}(\mathbb{F}_{miss})) + \dim((I_{metadata}(\mathbb{F}_{miss}))$

  Because the adaptive structural assignment (Axiom II) strictly increases the dimensional complexity of the relational metadata independently of the invariant constraints, the degrees of freedom df of the adversarial inference system evaluate to:

  $df = N_u - |C_{eq}| > 0$

  Proof Consequence. A mathematical system with strictly positive degrees of freedom ($df > 0$) is fundamentally underdetermined. Consequently, the invariant constraints C_{eq} cannot yield a unique deterministic solution for the missing fragments. The adversary is physically forced back into a state of combinatorial uncertainty. This structurally guarantees that the structural inference probability remains bounded strictly below certainty ($p_{inf} \ll 1$), preserving the Shannon Knowledge Restriction limit defined in Equation (18).
- **Lemma 9 (Decidability Constraint of Global Validation):**

  To prevent systemic undecidability and circumvent the Halting Problem, the evaluative functions governing the Global Veto are formally restricted to the class of total computable functions over finite domains.

  By Axiom I, the structural universe $\boldsymbol{S}$ and the active state space Σ are strictly finite. Let L_V be the formal logic or language utilized to express the systemic structural invariants (C) and relational consistency checks ($Cons_R$).

To ensure that the Boolean evaluation of the Global Validation Function $V_G(S(t))$ is always decidable, L_V is intentionally constrained to be Turing-incomplete. Specifically, L_V prohibits unbounded recursion and infinite operational loops.

Consequently, the validation of any candidate state S' maps strictly to a finite deterministic state automaton. Therefore, for every candidate state S', the evaluation of $V_G(S')$ is guaranteed to halt and converge to a binary output within bounded polynomial $O(n^k)$:

$$V_G(S') \in \{0, 1\} \ \forall S' \in \Sigma$$

Proof Consequence. Because V_G evaluates deterministically and universally halts, the State Transition Law (Equation 12a) is strictly decidable. The framework is thus immune to theoretical algorithmic paralysis, ensuring that the rejection of an invalid state is mathematically absolute and never subject to infinite deferral.

- **Lemma 10 (Semantic-Agnostic Topology and Entropy Separation):**

  To prevent the theoretical vulnerability of entropic overlap — whereby the sheer volume and structural shape of the topological metadata ($I_{metadata}$) inadvertently reveals the native semantic content ($I_{payload}$) at infinite fragmentation scales ($m \to \infty$) — the CNVS structural decomposition enforces strict statistical independence.

  Let the recursive fragmentation mapping $D_t(s)$ be fundamentally agnostic to the internal semantics of s. The decision boundaries defining the structural cuts and the subsequent verifier assignments are generated exclusively by the uniform cryptographic space Ξ defined in Axiom II.

  Because the physical fragmentation and relational topology are driven by a randomized oracle independent of the payload configuration, the topology acts as cryptographic white noise.

  Formally, the Shannon Mutual Information between the systemic metadata required for reconstruction and the native semantic payload is unconditionally zero:

  $$I(I_{metadata}(S) \ ; \ I_{payload}(S)) = 0$$

  Proof Consequence. Even as the system scales and the requisite metadata complexity diverges ($I_{metadata} \to \infty$) to maintain the graph connectivity, the structural "map" does not become isomorphic to the semantic "territory." Therefore, under a theoretical worst-case assumption where an adversary hypothetically compromises the system and gains access to the complete topological routing data, they would still extract exactly zero bits of useful information regarding the native payload. The epistemic restriction thus remains perfectly unbroken at any arbitrary fragmentation depth.

14. Principle of Knowledge Restriction

The CNVS framework does not assume Knowledge Restriction as a passive, static property, but rather enforces it as an active, structural consequence of adaptive fragmentation.

Under the Composite Density control mechanism defined in Chapter 8, any local accumulation of information triggers additional recursive fragmentation. Let $m(t) \triangleq \min_{s \notin Ter} |D(s)|$ denote the minimum active fragmentation depth imposed by the system at time t.

As the adaptive fragmentation increases, the verifier-accessible knowledge becomes strictly negligible relative to the total informational structure. Thus, we define the epistemic restriction via the following asymptotic limit:

(17)

$$\lim_{m(t) \to \infty} \mathcal{K}(v,f) / I_{global}(S(t)) = 0$$

To translate this structural bound into rigorous cryptographic terms, we apply Shannon Information Theory. Let X_s be the random variable representing the true internal semantic payload of a node s, and Y_v be the random variable representing the externally observed data by verifier v. The verifier's knowledge provides asymptotically zero information about the native semantic content. Formally, the Mutual Information $I(X_s ; Y_v)$ approaches zero, meaning there exists a vanishing sequence $\varepsilon_m \to 0$ as $m \to \infty$ such that:

(18)

$$I(X_s; Y_v) \leq \varepsilon_m$$

Equivalently, the Conditional Entropy of the internal payload, given the verifier's knowledge, remains strictly bounded near its absolute maximal uncertainty:

(19)

$$H(X_s \mid Y_v) \geq H(X_s) - \varepsilon_m$$

This proves that epistemic restriction is not an external assumption, but a mathematically inevitable consequence of density-controlled subdivision.

15. Probabilistic Security Model

We formalize the security bounds against an adversarial coalition attempting unauthorized state reconstruction.

Let:

- $\mathcal{V}$ be the finite set of total verifiers, with $Q = |\mathcal{V}|$.
- $\mathcal{V}_{adv} \subset \mathcal{V}$ be the adversarial colluding coalition.
- $k = |Ter(t)|$ be the total number of active terminal fragments.
- $q = |\mathcal{V}_{adv}| / Q$ be the probability of any single fragment being assigned to the adversarial coalition, assuming the uniform random assignment A_t defined in Axiom II.

Let $f_i \in Ter(t)$ be a terminal fragment. We define the normalized informational weight $\omega_i \in (0, 1]$ of the fragment relative to the maximal system entropy $H(S_{max})$ as:

(20)

$$\omega_i = I_{payload}(f_i) / H(S_{max})$$

Let $Y_i \in \{0, 1\}$ be the independent indicator random variable for adversarial acquisition:

- $Y_i = 1$ if f_i is assigned to a colluding verifier $v \in \mathcal{V}_{adv}$.
- $Y_i = 0$ otherwise.

The cumulative adversarial information acquired by the coalition $\mathcal{V}_{adv}$ is the random variable W:

(21)

$$W = \Sigma^{(k)}_{i=1} \; Y_i * \omega_i$$

Unauthorized systemic reconstruction is strictly defined to occur if and only if the accumulated information surpasses a critical reconstruction threshold $\tau \in (0, 1]$:

(22)

$$W \geq \tau$$

To evaluate the adversarial risk, let $\bar{\omega} = 1/k \, \Sigma^{(k)}_{i=1} \; \omega_i$ be the mean fragmental weight. The Expected Value of the adversarial informational acquisition $\mathbb{E}[W]$ evaluates to:

(23)

$\mathbb{E}[W] = \mathbb{E}[\Sigma^{(k)}_{i=1} Y_i * \omega_i] = \Sigma^{(k)}_{i=1} \mathbb{E}[Y_i]\omega_i = \Sigma^{(k)}_{i=1} q * \omega_i$

(24)

$\mathbb{E}[W] = q * k * \bar{\omega}$

This formal expectation proves that adversarial success is strictly bounded by the ratio of colluders q and the intentionally suppressed mean fragment weight $\bar{\omega}$.

16. Conditional Security Theorem

We establish the foundational condition under which adversarial reconstruction is probabilistically bounded. Let the absolute reconstruction threshold be defined as $\mathbb{T} = \tau k$, where $\tau \in (0, 1]$ represents the critical fraction of the total system capacity required to compromise the state.

Theorem 1 (Conditional Security)
If the strict inequality $q\, \bar{\omega} < \tau$ holds, the expected useful information acquired by the colluding coalition remains strictly below the threshold required for unauthorized state reconstruction.

Proof

From Equation (24), the expected adversarial information is $\mathbb{E}[W] = q * k * \bar{\omega}$.
By algebraic substitution of the theorem's condition ($q\, \bar{\omega} < \tau$) scaled by the fragment count k:

(25)

$\mathbb{E}[W] < \tau k$

Under the idealized approximation of independent probabilistic assignment A_t, the probability of a successful adversarial reconstruction satisfies the asymptotic bound:

(26)

$\mathrm{Lim}_{k \to \infty} \mathbb{P}(W \geq \tau k) = 0$

This demonstrates that as long as the threshold exceeds the expected acquisition, the system is conditionally secure.

17. Exponential Concentration Bound

To prove the rate of convergence for Equation (26), we quantify the probability distribution of the adversarial information W using measure concentration inequalities.

Theorem 2 (Hoeffding Concentration Bound). Given k independent random assignments, the probability of unauthorized reconstruction decreases exponentially as the number of active terminal fragments k increases, provided $q\,\bar{\omega} < \tau$.

Proof.

The total adversarial information $W = \Sigma^{(k)}_{i=1}\ Y_i * \omega_i$ is the sum of independent random variables. Since $Y_i \in \{0, 1\}$, each weighted contribution $Y_i * \omega_i$ is strictly bounded within the interval $[0, \omega_i]$.

We seek to bound the probability that W exceeds the threshold τk. We rewrite this deviation from the mean $\mathbb{E}[W] = q\,k\,\bar{\omega}$ as:

$\mathbb{P}(W \geq \tau k) = \boldsymbol{\mathcal{P}}(W - \mathbb{E}[W] \geq k(\tau - q\,\bar{\omega}))$

By applying Hoeffding's Inequality for bounded independent variables, we obtain the exponential concentration bound:

(27)

$\mathbb{P}(W \geq \tau k) \leq \exp\left(- (2k^2 (\tau - q\,\bar{\omega})^2) / (\Sigma^{(k)}_{i=1}\ \omega_i^2) \right)$

Since $\tau - q\,\bar{\omega} > 0$ by the hypothesis of Conditional Security, the exponent is strictly negative. Since the numerator grows quadratically ($O(k^2)$) while the denominator grows linearly ($O(k)$), the overall magnitude of the exponent scales linearly with k, forcing a strictly exponential probability decay.

18. Emergent Security Scaling Theorem

We finalize the proof of thermodynamic security by demonstrating the absolute asymptotic limit of the concentration bound.

Theorem 3 (Asymptotic Emergent Security)

Let $m \in \mathbb{N}$ denote the global state fragmentation scale such that as $m \to \infty$, the number of terminal fragments $k(m) \to \infty$. If the assignment condition $q\,\bar{\omega} < \tau$ is maintained, the probability of systemic compromise converges to zero.

Proof.

We analyze the exponent of the Hoeffding bound derived in Equation (27). Since the normalized informational weights ω_i are bounded by definition (ω_i ≤ 1), the sum of their squares is bounded by the fragment count: $\Sigma^{(k)}_{i=1} \omega_i^2 \leq k$.

Substituting this upper bound into the denominator of Equation (27) yields a strictly looser, yet linearly diverging, upper bound for the probability:

$\mathbb{P}(W \geq \tau k(m)) \leq \exp (- (2k(m)^2 (\tau - q\ \bar{\omega})^2)/ k(m)) = \exp (-2k(m) (\tau - q\ \bar{\omega})^2)$

Let the constant gap be $\Delta = (\tau - q\ \bar{\omega})^2 > 0$. The bound simplifies to:

(28)

$\mathbb{P}(W \geq \tau k(m)) \leq \exp (-2\Delta\ \ k(m))$

As the fragmentation scale expands ($k(m) \to \infty$), the linear term $-2\Delta\ \ k(m)$ diverges deterministically to $-\infty$.

Consequently, the exponential function evaluates to zero:

$\lim_{m \to \infty} \mathbb{P}(W \geq \tau k(m)) = 0$

Thus, systemic security emerges inherently from the topological scaling of fragmentation, neutralizing the adversarial coalition irrespective of their finite computational capacity.

18a. Generalized Emergent Security Scaling under Dependent Collusion

The independent-assignment concentration analysis (Chapters 16-18) provides an idealized asymptotic bound. We now generalize the model to evaluate dependent compromise events, where an adversarial coalition attempts cryptanalytic inference across multiple fragments.

Let:

- $q \in [0, 1)$ denote the probability that a critical fragment is directly assigned to colluding verifiers (as defined in Axiom II).
- $p_{inf} \in [0, 1)$ denote the probability of structural inference — the likelihood that a fragment not directly controlled can be compromised via cryptanalytic derivation from already compromised fragments.
- $m \in \mathbb{N}$ denote the strictly positive number of critical low-inference fragments required for unauthorized systemic reconstruction.

We formally define the single-fragment composite compromise probability p_{comp} via the Law of Total Probability:

(29)

$p_{comp} \triangleq q + (1 - q)\, p_{inf}$

Let c_{coll_i} denote the adversarial event that the i-th critical fragment is successfully compromised. To formalize the conditional dependence between compromise events, we establish the following bounding lemma.

Lemma 7 (Structural Inference Bound). By the Principle of Knowledge Restriction defined in Equation (19), the mutual information between structurally isolated fragments is asymptotically negligible. Consequently, the adversarial knowledge gained from compromising i-1 fragments yields no deterministic decryption path for the i-th fragment. The conditional compromise probability is therefore strictly bounded by the composite threshold:

(30)

$\mathbb{P}(c_{coll\_i} \mid c_{coll\_1}, \ldots., c_{\_coll\_i-1}) \leq p_{comp}$

with $0 \leq p_{comp} < 1$.

Theorem 4 (Generalized Emergent Security Scaling)

Under the constraints of the Structural Inference Bound, the probability of an unauthorized full state reconstruction, denoted as Rec^*, satisfies the exponential decay inequality:*

(31)

$\mathbb{P}(Rec^*) \leq (p_{comp})^{\wedge m}$

Proof.

Unauthorized full reconstruction strictly requires the simultaneous compromise of all m critical low-inference fragments. In set-theoretic terms, the reconstruction event is a subset of the intersection of all compromise events:

$Rec^* \subseteq \cap^m_{i=1} c_{coll\_i}$

Therefore, by the monotonicity of probability measures:

$\mathbb{P}(Rec^*) \leq \mathbb{P}(\cap^m_{i=1} c_{coll\_i})$

Applying the general chain rule of probability:

$$\mathbb{P}(\cap^m_{i=1} c_{coll\_i}) = \prod^m_{i=1} \mathbb{P}(c_{coll\_i} \mid c_{coll\_1}, \ldots., c_{coll\_i-1})$$

Substituting the strict upper bound derived in Lemma 7 (Equation 30) into the product series:

(32)

$\mathbb{P}(Rec^{\wedge *}) \leq \prod^{m}_{i=1} p_{comp} = p_{comp}^{m}$

This demonstrates that systemic security scales exponentially against dependent collusion, provided the underlying composite probability remains strictly sub-unitary.

Corollary 4.1 (Asymptotic Dependent Security)

Because $0 \leq p_{comp} < 1$, the geometric sequence $(p_{comp})^m$ converges to zero. Therefore, as the system scale expands and the requisite critical fragments increase:

$\lim_{m \to \infty} \mathbb{P}(Rec^*) = 0$

Corollary 4.2 (Non-Inferable Limit)

If the systemic metadata overhead ensures perfect cryptanalytic isolation such that the inference probability vanishes ($p_{inf} = 0$), then $p_{comp} = q$. The generalized security scaling converges perfectly to the idealized probabilistic boundary:

$\mathbb{P}(Rec^*) \leq q^m$

Systemic Design Formula: Minimum Critical Fragmentation

To operationalize the security model, the framework must determine the minimum decomposition depth required to satisfy a specific security target. Given a maximum tolerable target attack probability $\eta \in (0, 1)$, we seek the minimum number of critical low-inference fragments m_min such that:

$\mathbb{P}(Rec^*) \leq \eta$

By substituting the exponential bound derived in Theorem 4:

$p_{comp}^{m} \leq \eta$

Taking the natural logarithm of both sides:

$m \ln(p_{comp}) \leq \ln(\eta)$

Since $\eta \in (0, 1)$ and $p_{comp} \in (0, 1)$, both logarithmic terms are strictly negative. Consequently, dividing both sides by ln(p_comp) reverses the inequality direction:

$m \geq \ln(\eta) / \ln(p_{comp})$

Because the fragmentation depth m must be a discrete natural number ($m \in \mathbb{N}$), we define the exact threshold using the ceiling function:

(33)

$m_{min} = \lceil \ln(\eta) / \ln(p_{comp}) \rceil$

This formula deterministically translates any theoretical cryptographic security requirement (η) into a precise topological constraint (m_{min}).

Remark 18a.1 (Semantic lower bound and practical scaling)

The asymptotic expression $m \to \infty$ utilized in Corollary 4.1 represents an idealized mathematical limit. In practical CNVS deployments, fragmentation cannot proceed indefinitely.

As formalized in Chapter 6, terminal fragments must preserve a sufficient semantic payload to allow honest external verifiers to perform their assigned local evaluation tasks without logical undecidability.

Therefore, the effective fragmentation depth is strictly constrained by the semantic lower bound:

(34)

$\forall f \in \mathrm{Ter}(t),\ \ I_{payload}(f) \geq I_{min}$

where I_{min} denotes the absolute minimum informational quantum required for task comprehension.

Consequently, the practical system design problem is not to maximize fragmentation to infinity, but to solve a Bounded Optimization Problem:

Maximize epistemic restriction (by increasing m) subject to the constraint of local semantic verifiability.

Let m_{max} denote the theoretical maximum number of fragments derivable from a root element s before the payload of any resulting terminal fragment drops below I_{min}.

A specific CNVS configuration is considered structurally and cryptographically feasible if and only if the maximum semantically permissible fragmentation equals or exceeds the minimum security requirement:

(35)

$m_{max} \geq m_{min}$

This feasibility condition formally guarantees that the system can achieve the target adversarial resistance η without sacrificing the operational computability of the global state.

Remark 18a.2 (Entropic interpretation of inference probability)

The structural inference probability p_{inf} is not a static scalar, but dynamically depends on the residual uncertainty of the global state.

Let W denote the cumulative weighted adversarial information controlled by the coalition, as defined in Equation (21).
As W increases, the residual uncertainty of the global state decreases, monotonically increasing the probability of inferring non-controlled critical fragments via cryptanalysis.

We formalize this by defining p_inf as a non-decreasing probability function of the accumulated knowledge:

$$p_{inf} = \nu(W)$$

where $\nu : [0, 1] \to [0, 1)$ maps the normalized adversarial information to the inference likelihood.

Substituting this into Equation (29), the generalized compromise probability becomes dynamically coupled to the entropy:

(36)

$$p_{comp} = q + (1 - q)\, \nu(\mathbb{E}(W))$$

Consequently, the generalized exponential bound (Theorem 4) updates to:

(37)

$$\mathbb{P}(\mathrm{Rec}^{*}) \leq [q + (1 - q)\, \nu(\mathbb{E}(W))]^{m}$$

This explicitly unifies the dependent-collusion model with the thermodynamic interpretation of fragment utility, formally bridging Shannon entropy with systemic cryptanalysis.

Remark 18a.3 (Worst-case leakage resilience)

The emergent security scaling demonstrated in Theorems 3 and 4 remains mathematically sound even under a worst-case catastrophic local leakage scenario.
Assume a boundary condition where the local convergence function (Conv_{σ}) fails to project a minimized semantic view, granting the colluding verifiers absolute local knowledge of their assigned fragments. Formally, the Knowledge Restriction bound (Equation 17) collapses locally, yielding:

$$\mathcal{K}(v,f) = I_{total}(f), \quad \forall v \in \mathcal{V}$$

Even under this worst-case assumption, as long as the composite threshold $p_{comp} < 1$, the asymptotic limit established in Corollary 4.1 holds absolute:

$\lim_{m \to \infty} \mathbb{P}(Rec^*) = 0$

This proves that the local verification mechanism (V_L) and the thermodynamic fragmentation topology (D) act as orthogonal defensive layers. The emergent macroscopic security remains valid independently of localized sub-fragmentary leakage.

Theorem Implications:

This result formally generalizes the idealized independent-assignment model. It proves that exponential security scaling against coordinated collusion is guaranteed, provided that the inference probability of any single fragment remains strictly bounded by a constant smaller than one.

19. Combinatorial Entropy and Systemic Scaling interpretation

The structural growth of the CNVS framework mathematically guarantees an explosion in epistemic restriction. As the system scales, unauthorized reconstruction transitions from computationally hard to exponentially improbable.

This security is an emergent property of the topological state space. In the uniform random assignment mapping $A_t : Ter(t) \times \Xi \to \mathcal{V}$ (Axiom II), the number of possible routing configurations grows factorially with the number of terminal fragments k.

We define the Combinatorial Entropy of the system assignment as H_{comb}(measured in Sbits):

(38)

$H_{comb}(k) = \log_2(k!)$

By applying Stirling's approximation ($\ln(k!) \approx k \ln k - k$), it is evident that the combinatorial entropy scales at a bounding rate of $O(k \log k)$.

Therefore, taking the limit as the system expands:

$\lim_{m \to \infty} H_{comb}(k) = \infty$

This factorial divergence formally dictates that the structural uncertainty faced by an external adversary grows vastly faster than any linear accumulation of fragmental information, physically precluding brute-force graph reconstruction.

19a. Corollary: Global Veto Property

If at least one critical fragment essential to the evaluation of a global invariant $c_i \in C$ remains outside adversarial control, the deterministic reconstruction of a globally valid state is impossible. Any incorrect adversarial estimation of the missing structural or relational contribution will force the invariant to fail, triggering the absolute Global Veto:

$V_G(S') = 0$

and resulting in the immediate topological rejection of the candidate state S'.

19b. Critical Fragment Principle

The security of the CNVS framework does not scale linearly with the percentage of Verifiers under adversarial control, but rather hinges on the absolute combinatorial uncertainty of the uncompromised fragments.

We formally define a fragment $f^* \in Ter(t)$ as Essential and Non-Inferable if its internal metadata and relational topology cannot be deterministically computed from the information vector W held by the adversarial coalition $\mathcal{V}_{adv}$.

Let $F_C \subset Ter(t)$ be the subset of fragments assigned to the coalition. If there exists at least one uncompromised critical fragment ($f^* \notin F_C$), the adversary is forced to correctly guess its state to bypass the Global Veto.

By the principles of Information Theory, the probability of successfully reconstructing the global state becomes strictly bounded by the combinatorial entropy of the missing fragment's information measure $I_{total}(f^*)$:

(39)

$\mathbb{P}(Rec^*) \leq 2^{-Itotal(f^*)}$

Conclusion: Extensive collusion does not imply linear success. The existence of a single indispensable fragment outside adversarial control forces the probability of unauthorized reconstruction to drop exponentially, irrespective of the sheer volume of compromised nodes.

Remark 19.1 (The "N-1" Byzantine Example)

Suppose a CNVS state encodes an execution graph of k = 100 financial allocation orders, strictly bound by a global conservation invariant ($c_{conservation} \in C$) and hidden relational dependencies ($Cons_R$). Assume an adversarial coalition controls 99 out of the 100 terminal fragments. While they possess 99% of the local observations, they lack the missing relational assignment and the specific invariant parameters encoded in the 100_{th} fragment. Because the local observations do not contain the native global payload (as proven by the Knowledge Restriction limit), the missing data is not algebraically inferable. An adversarial attempt to reconstruct the state graph using guessed parameters will violate c_conservation or $Cons_R$, yielding $V_G(S) = 0$. As the systemic metadata complexity $I_{metadata}$ increases, the probability of correctly fabricating the missing 1% drops exponentially, preserving the 100% integrity of the system.

Remark 19.2 (Reactor Example)

Semantic Decoupling and the Construction Site Paradox

To fully grasp the ontology of a "terminal fragment" within the CNVS framework, it is useful to employ a physical analogy: the validated construction of a classified nuclear reactor (the Global Payload).

In a traditional distributed system (e.g., data sharding), fragmentation would be equivalent to taking the engineering blueprint of the reactor, tearing it into a thousand pieces, and distributing those pieces to the workers. Even though the information is fragmented, the workers possess literal fractions of the semantic Payload. If they collude and aggregate a sufficient number of pieces, they can infer the existence and nature of the reactor.

The CNVS enforces a radical ontological decoupling. The system does not distribute "pieces of the blueprint," but rather autonomous measurement directives ($I_{metadata}$).

The worker (the external verifier) is never provided with the expected data. The system does not ask: "Can you confirm this screw is exactly 5 millimeters?" (which would leak part of the state expectation). Instead, the system issues an open, agnostic directive: "Measure the length and head diameter of this specific screw and report the numerical value."

The verifier acts as a pure physical oracle. The worker possesses sufficient local semantics to execute the task (they know what a screw is and how to use a caliper), but the factual data flows entirely from the physical world into the system. Because the relational topology linking that individual screw to the overarching design is cryptographically obscured, the Mutual Information between the local diameter measurement and the global semantic payload (the construction of a nuclear reactor) is physically and mathematically zero.

Conclusion

The adversary faces three possible scenarios. Let's analyze them.
Scenario 1: He has the 99 Payloads, but no Topology.
The adversary controls 99 out of 100 verifiers. He has 99 "real physical measurements." But not having the map (the Topology Ξ), he can't know how these measurements fit together. He has 99 random pieces of information, but not knowing the Invariant equation, he can't deduce the last missing piece of information.
The system is safe.

Scenario 2: He has the Topology, but no Payloads.
The adversary steals the map from the server, but he doesn't have control over the physical verifiers or their assignment functions. In this case, Mutual Information is absolute zero: seeing the routing map ($I_{metadata}$) tells him nothing about what the verifiers are actually measuring ($I_{payload}$).
The system is safe.
Scenario 3: Has the 99 Payloads and also has the Topology.
The adversary has stolen the map and controls 99% of the network. He knows the data and how it connects.
Theoretically, the CNVS could demonstrate resilience even in this scenario.
In a traditional cryptosystem (such as Shamir Secret Sharing), Scenario 3 means a complete Game Over.
If the adversary has 99 pieces and knows the mathematical structure of the polynomial (the topology), he can calculate the missing 100_{th} piece.

In the CNVS, there are two lines of structural defense that can still provide some resilience:
1. The Measurement Grid (Physical Decoupling):
The fragment is not a chopped-up payload piece, but a measurement grid. Even if the adversary has the complete map and 99 measurements, the 100_{th} measurement resides in the real world, in the physical domain of the 100_{th} honest verifier. The topological map can provide information about the measurement grid of the 100_{th} node, but it cannot provide the outcome of the physical observation that that node must perform, because it does not belong to it, being assigned to a non-colluding verifier.
2. For Lemma 8 (Topological Underdetermination):
This lemma mathematically addresses the discussion regarding the Degrees of Freedom (df) and shows that they always remain positive. Even by combining the Map and the 99 Payloads, the Global Invariant equation (C_{eq}) remains underdetermined.
The adversary does not have enough mathematical constraints to solve the equation and derive the 100_{th} payload with certainty.
Conclusion

Theoretically, there is an intrinsic resistance structure within the apparatus, the "Shannon Wall" (Lemma 10): the map alone does not allow for inferring the payload. This is because the decomposition into CNVSs does not occur as a geometric cut, but rather constitutes an assignment of oracular tasks (the measurement grids) combined with a randomized and irreducible relational graph.

20. Scientific Positioning

The CNVS architecture synthesizes principles from Shannon Information Theory, probabilistic combinatorics, structural graph theory, and distributed formal verification into a unified mathematical framework.

Crucially, CNVS diverges structurally from traditional threshold cryptography (e.g., Shamir's Secret Sharing). In standard threshold models, the fragment (the "share") intrinsically contains a mathematical projection of the plaintext payload. In the CNVS framework, the verifier is never provided with a data share containing the internal semantic payload. Instead, the verifier receives strictly bounded metadata dictating a localized measurement task, subsequently producing observational data used solely for convergence verification. This epistemic decoupling ensures that systemic security scales not by complex cryptographic key-splitting, but through the thermodynamic limits of structural graph reconstruction.

21. Open Problems

While this paper establishes the foundational formal proofs of the CNVS architecture, several advanced vectors require further formalization:

1. Cryptographic Reductions: Formal mapping of the local knowledge restriction bounds to standard hard computational problems (e.g., Learning With Errors (LWE) or Discrete Logarithm).
2. Information-Theoretic Lower Bounds: Derivation of the absolute optimal adversarial allocation strategies and the corresponding tightest bounds for $I_{metadata}$ expansion.
3. Empirical Validation: Extensive stress-testing of the asymptotic bounds ($\mathcal{P}(Rec^*) \to 0$) via large-scale Monte Carlo systemic simulations under dynamically shifting adversarial assignment (A_σ).

Appendix A. Minimal Formal Definition

A structure

$M = (\mathcal{S}, E, R, C, D, A, V_L, V_G, Rec, K_F)$

is a Closed Native Verification System if and only if:

1. $\mathcal{S}$ is a typed finite-binary universe (Def.1).
2. Terminal membership requires local admissibility, while State validity requires global verification (V_G).
3. Global validity requires relation consistency and invariants.
4. Decomposition is finite, recursive, and reconstructible.
5. Verifier knowledge is strictly restricted (Eq.19).
6. Terminal assignments are randomized.
7. Invalid elements and states are rejected.
8. In this document, the symbols V_L and V_G denote respectively the Local Verification Function (V_{Local}) and the Global Verification Function (V_{Global}) introduced in the extended CNVS theory.
9. K_F denotes Fragment-Level Knowledge, i.e. the semantic knowledge associated with a terminal Fragment. In the extended CNVS theory, K_F belongs to a broader hierarchy of knowledge levels, but only the fragment-level representation is required for the compact formal formulation.

Appendix B. Formula Correspondence with the Extended CNVS Theory

The present Formal Theory is a compact mathematical synthesis of the complete CNVS framework presented in:

The Theory of Closed Native Verification Systems (CNVS) v5.5.

For traceability purposes, the following table indicates the correspondence between the formulas introduced in this document and their principal counterparts in the extended theory.

Formal Theory	Extended Theory	Formal Theory	Extended Theory
(1)	Ext.4	(21)	Ext.114
(2)	Ext.5	(22)	Sec.16.2

Formal Theory	Extended Theory	Formal Theory	Extended Theory
(3)	Ext.6–7	(23)	Ext.113
(4)	Ext.9	(24)	Ext.114
(5)	Sec.5.8.1	(25)	Ext.103
(6)	Ext.11	(26)	Sec.16.2
(7)	Ext.14	(27)	Sec.16.2
(8)(8a)	Ext.57–61	(28)	Sec.16.2
(9)	Ext.57	(29)	Ext.115
(10)	Ext.65	(30)	Ext.116
(11)	Ext.66	(31)	Ext.117
(12)	Ext.68	(32)	Ext.118
(13)	Ext.62	(33)	Ext.120
(14)	App.A.4-B.4	(34)	Ext.119
(15)	Ext.15,55,56,62	(35)	Sec.17.2
(16)	Ext.27–28	(36)	Ext.121
(17)(17a)	Ext.57–61	(37)	Ext.122
(18)	App.A.13.1	(38)	Ext.123
(19)	App.A.13.2	(39)	Ext.124
(20)	Ext.112		

The compact formulation intentionally omits a number of intermediate derivations, operational constraints, entropy analyses, and implementation-oriented formal developments contained in the extended CNVS theory. Whenever discrepancies arise, the extended theory shall be considered the authoritative mathematical reference.

AI-Assistance Disclosure

The author acknowledges the use of AI tools for editorial refinement, language optimization, formatting support, and consistency checking. The conceptual architecture, axiomatic structure, mathematical definitions, and theoretical claims of the CNVS framework were determined by the author, who assumes full scientific responsibility for the content of this manuscript.

END DOCUMENT

Author:

Massimo Comitato, Milano (MI),

Italy,   07.06.2026